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RESEARCH ARTICLE

Team informational resources, information elaboration, and team innovation: Diversity mindset moderating functional diversity and boundary spanning scouting effects

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Abstract

The knowledge integration perspective on team innovation holds that information elaboration – the exchange, discussion, and integration of task-relevant information and perspectives – is the core team process driving team innovation. Factors reflecting the informational resources the team can draw on through information elaboration therefore are important influences on team innovation. In this respect, team innovation research points to team functional diversity and to team boundary spanning scouting to acquire information from outside the team. Team innovation research also makes clear that informational resources (as reflected in functional diversity and boundary spanning scouting) do not guarantee team information elaboration, and that identifying moderation in this relationship is particularly valuable. Building on this state of the science, we focus on the moderating role of the team diversity mindset – members' shared understanding of the importance of information elaboration for team performance – in the relationships of team functional diversity and boundary spanning scouting with information elaboration and team innovation. A multi-wave and multi-source survey of $N=215$ teams involved in knowledge work in various Chinese organizations supported our research model for team boundary spanning scouting but not for team functional diversity.

KEYWORDS

boundary spanning scouting, diversity mindset, functional diversity, information elaboration, team innovation

Innovation, the generation and implementation of novel and useful outcomes, is of critical importance to organizations (Shalley et al., 2015). Teams are a key source of innovation in organizations, making team innovation a core focus for innovation research (Hülsheger et al., 2009). An integrative review identified the knowledge integration perspective as the dominant approach in team innovation research (van Knippenberg, 2017). The knowledge integration perspective recognizes team information elaboration – the exchange, discussion, and integration of task-relevant information and perspectives (van Knippenberg et al., 2004) – as the core team process driving team innovation. Information elaboration draws on the team's informational resources – the knowledge and perspectives members bring to the team. A direct implication of this stream is factors that contribute to team informational resources can provide an important stimulus for information elaboration and team innovation.

In recognition of the importance of team informational resources, team innovation research has placed particular emphasis on functional background diversity as an influence on team innovation (Hülsheger et al., 2009; van Knippenberg, 2017). Members with different functional backgrounds bring different expertise, experience, and perspectives to the team, resulting in greater and more diverse informational resources for more functionally diverse teams. Meta-analytic evidence supports the conclusion that team functional diversity is positively related to team innovation (Hülsheger et al., 2009; van Dijk et al., 2012). The evidence also indicates that functional diversity is not guaranteed to result in information elaboration and team innovation and that moderating influences that stimulate (rather than disrupt) information elaboration are key in this respect (van Dijk et al., 2012; van Knippenberg & Hoever, 2021).

Although functional diversity concerns informational resources brought to the team through team composition, team innovation research also recognizes that team members may engage in boundary spanning – interact with others outside of team boundaries – to scout for information to bring to the team (boundary spanning is not limited to scouting for information, Ancona & Caldwell, 1992, but from the knowledge integration perspective we focus on boundary spanning scouting, because this is the element that concerns team informational resources). Boundary spanning scouting received less attention in team innovation research than team functional diversity, but enough to conclude that it is a positive influence on team innovation (Ancona & Caldwell, 1992; Howell & Shea, 2006; Hülsheger et al., 2009). Like team functional diversity, the influence of boundary spanning scouting may be understood from the perspective that it gives the team greater and more diverse informational resources (van Knippenberg, 2017). To our knowledge, team innovation research has not considered the relationship between boundary spanning scouting and information elaboration, but the logic that holds for team functional diversity should apply here: Informational resources (i.e., from boundary spanning scouting) cannot be equated to using these resources through information elaboration (De Dreu et al., 2008; van Knippenberg et al., 2004). In developing the knowledge integration perspective, an important issue to address therefore is moderation in the relationship of boundary spanning scouting with information elaboration and, mediated by elaboration, team innovation.

We address this issue from the perspective that there is added value in developing theory about moderation in the relationship of boundary spanning scouting with elaboration and innovation when that theory is well-aligned with the state of the science in the moderated relationship of team diversity with elaboration and innovation (cf. Cronin et al.'s, 2021, argument regarding the importance of theory that bridges research streams). To build towards theory that integrates team diversity and team boundary spanning perspectives on team innovation, we address moderation for functional diversity as well as boundary spanning scouting by recognizing the communalities between team functional diversity and boundary spanning scouting in (i) both reflecting team informational resources that (ii) can stimulate team innovation through information elaboration, which (iii) is contingent on moderating influences. Recognizing these communalities allows us to draw on theory in team diversity that emphasizes the core role of information elaboration and its contingencies (van Knippenberg et al., 2004, 2013) and that is well-aligned with the knowledge integration perspective on team innovation (van Knippenberg, 2017; van Knippenberg & Hoever, 2021). Specifically, it allows us to understand the influence of team functional diversity and boundary spanning scouting through the same mediating process (information elaboration) and through the same moderating influence in the relationship between factors reflecting team informational resources (diversity, boundary spanning scouting) and information elaboration.

In considering such moderation, we focus on team diversity mindset (van Knippenberg et al., 2013; van Knippenberg & van Ginkel, 2022). Diversity mindset is a form of team cognition (mental representations of the team and teamwork; Salas & Fiore, 2004) specific to team diversity. Following Newell and Simon (1972), who defined such mental representations as people's understanding of the actions required to achieve a particular goal, van Knippenberg et al. (2013) conceptualized diversity mindset as team members' understanding that information elaboration is required to realize the performance benefits of team diversity. Diversity mindset theory (van Knippenberg et al., 2013) builds on the categorization-elaboration model, the source of the team information elaboration concept (van Knippenberg et al., 2004). In doing so, diversity mindset theory recognizes that information elaboration is the core mediating process in the diversity–performance relationship as well as that elaboration does not automatically follow from team diversity (cf. the knowledge integration perspective on team innovation). Diversity mindset theory advances diversity mindset as an important moderator in this respect, because of the *procedural knowledge* captured by team cognition. Procedural knowledge refers to members' understanding of the actions required to achieve team goals and thus is important in guiding members' engagement in teamwork (Salas & Fiore, 2004; van Knippenberg et al., 2013). Diversity mindset is particularly relevant to the relationship between team informational resources and information elaboration, because it reflects members' understanding that elaboration is required to realize the performance benefits of the team's informational resources (i.e., diversity mindset recognizes that for any diversity attribute, to create value the focus should be on the informational diversity associated with the attribute; van Knippenberg et al., 2013). In this sense, as per van Knippenberg et al. (2013), diversity mindset provides more precision in capturing moderating influences in the relationship between team informational resources and information elaboration than the team climate perspectives (which also concerns shared perceptions) found in team innovation research that neither concerns elaboration nor procedural knowledge (cf. van Knippenberg, 2017).

Application of diversity mindset theory allows us to predict that team diversity mindset moderates the relationship of team functional diversity with information elaboration and, mediated by elaboration, team innovation. Although identifying this moderated mediation for functional diversity is a more modest contribution within the state of the science, it is important in setting the stage for the primary integrative contribution of our study – the proposition that this moderated mediation logic also holds for boundary spanning scouting. Extending diversity mindset theory and achieving integration between the study of team diversity and team boundary spanning scouting in team innovation, we propose that diversity mindset moderates the relationship of boundary spanning scouting with elaboration and innovation. The rationale here is that at root diversity mindset reflects an understanding of team diversity as an informational resource (van Knippenberg et al., 2013) and thus focuses team members' attention not on their surface differences but on uncovering and using underlying informational diversity. Boundary spanning scouting adds to team informational diversity by bringing new information, knowledge, and perspectives to the team. Thus, we can expect teams to capitalize on these resources through elaboration more with higher diversity mindset. Figure 1 captures our research model.

Our study contributes to the development of the knowledge integration perspective on team innovation (van Knippenberg, 2017). We develop this perspective by recognizing the communalities between team functional diversity and team boundary spanning scouting. Both provide the team with diverse informational resources and both can positively influence team innovation through information elaboration. For neither, however, elaboration will automatically follow; diversity mindset, with procedural knowledge of information elaboration at its core, can be proposed to be a moderating influence in their relationships with elaboration. Recognition of these communalities allows us to link functional diversity and boundary spanning scouting to team innovation through the same mediating and moderating influences. This is an important step in bridging the different streams of research on diversity and boundary spanning to build towards more integrative theory, which is key to advancing science (Cronin et al., 2021).

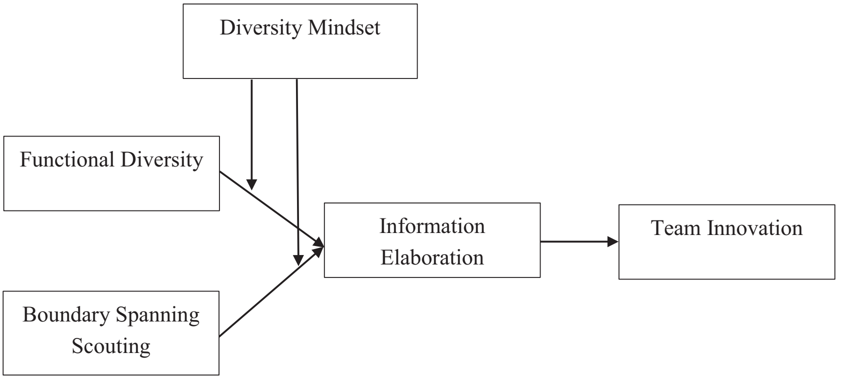

FIGURE 1 Research model.

THEORETICAL BACKGROUND AND HYPOTHESES

The generation of useful novelty is core to team innovation (Anderson et al., 2014). Such novelty is generated through the integration of diverse information and perspectives into new insights (Hargadon & Bechky, 2006). Teams are particularly suited for innovation, because different people know, think, and see different things. As a result, teams can draw on diverse knowledge, information, and perspectives as a basis to generate novel outcomes. The knowledge integration perspective on team innovation thus suggests that in understanding team innovation, it is important to consider factors that constitute team informational resources.

The factor that has been most studied in this respect is team functional background diversity (van Knippenberg & Hoever, 2021). Functional diversity captures the extent to which team members have different functional backgrounds and can be assumed to have different job-relevant knowledge, information, and perspectives. Greater functional diversity provides the team with a greater pool of diverse informational resources to draw on and thus with a better basis to achieve innovative outcomes through knowledge integration. Functional diversity should therefore be a positive influence on team innovation. Meta-analyses have established that on average it is (Hülsheger et al., 2009; van Dijk et al., 2012). Meta-analysis also shows, however, that there is significant heterogeneity in these effects and that they range from negative to positive (van Dijk et al., 2012). This puts a premium on developing moderation theory to capture under what conditions team innovation benefits from functional diversity.

Complementing this team composition perspective, the notion that team innovation benefits from diverse informational resources is also found in research on team boundary spanning to bring new information to the team (Phelps et al., 2012; Schotter et al., 2017; boundary spanning can also serve other functions (Ancona & Caldwell, 1992) but from the knowledge integration perspective we only consider boundary spanning in the form of scouting for external information). Boundary spanning scouting adds to team informational resources and may thus feed into team information elaboration. Not surprisingly, then, boundary spanning is a positive influence on team innovation (Ancona & Caldwell, 1992; Howell & Shea, 2006).

We propose that there is an important communality between boundary spanning scouting and functional diversity: Bringing informational resources to the team. Because informational resources do not automatically result in information elaboration, however (De Dreu et al., 2008), there is no reason to expect that information elaboration automatically follows from boundary spanning scouting. Instead, we may expect that the effect of boundary spanning scouting on elaboration (and thus on innovation) is moderated by factors that stimulate team information elaboration, just like the effects of functional diversity. In recognition of this, we consider the moderating role of diversity mindset as the factor that is most explicitly conceived in terms of a focus on information elaboration (van Knippenberg et al., 2013).

In the following we develop hypotheses about how diversity mindset moderates both the relationship of team functional diversity with elaboration and innovation and the relationship of team boundary spanning with elaboration and innovation.

Functional diversity and diversity mindset

Research on team information elaboration shows that teams may often insufficiently recognize that their team performance benefits from information elaboration and instead prioritize, for instance, identifying what team members agree on (i.e., as opposed to seeing diverging viewpoints as input for elaboration; van Ginkel & van Knippenberg, 2008, 2009). Because teams can only benefit from the underlying informational diversity associated with team diversity attributes to the extent that this diversity of information and perspectives is exchanged, discussed, and integrated (van Knippenberg et al., 2004), such insufficient focus on information elaboration is an important obstacle to realizing the performance benefits of diversity (Ely & Thomas, 2001; Kooij-de Bode et al., 2008). The concept of diversity mindset was proposed to address this very issue (van Knippenberg et al., 2013).

In van Knippenberg et al.'s (2013) diversity mindset theory, diversity mindset is conceptualized as team cognition reflecting an understanding that team performance on task with a more explorative, non-routine focus (i.e., like innovation) can benefit from team diversity as an informational resource and that to realize these benefits the team needs to engage in information elaboration. Diversity mindset is understood to be specific to the team; it reflects members' understanding of the value of diversity for their team specifically (rather than generically for teams). Diversity mindset is also specific to diversity attribute; an understanding of the value of diversity in one attribute (e.g., functional background) does not automatically imply an understanding of the value of diversity in another attribute (e.g., age) or diversity in general.

Diversity mindset theory also outlines that diversity mindset, like other team cognition (Salas & Fiore, 2004), derives its influences from being shared cognition (i.e., similarity among team members in diversity mindset). Team cognition such as diversity mindset is an important influence on team process, because team cognition reflects members' understanding of how to operate as a team to achieve team goals. That is, team cognition – and in particular those aspects of team cognition that concern teamwork processes like diversity mindset – guides team task performance (Salas & Fiore, 2004; van Knippenberg et al., 2013). Shared cognition is important here, because it ensures that members' actions are aligned (i.e., because they are guided by similar understanding of team process) and because such aligned action reinforce acting on one's understanding of team process (Morgeson & Hoffman, 1999).

For team innovation to benefit from team functional diversity, information elaboration is the key mediating mechanism (van Knippenberg & Hoever, 2021). Thus, diversity mindset (concerning functional diversity) can be proposed to play an important moderating role in the relationship between functional diversity and elaboration, because it focuses team members on the importance of information elaboration to realize the performance benefits of functional diversity. Because diversity mindset concerns not only members' focus on information elaboration but also their understanding that information elaboration is particularly valuable for explorative performance (e.g., team innovation; as opposed to more routine performance; van Knippenberg et al., 2013), diversity mindset should be especially relevant when the outcome of interest is team innovation. Thus, higher diversity mindset can be expected to motivate more team information elaboration to draw on the team informational resources provided by team functional diversity. In doing so, diversity mindset can be proposed to strengthen a positive relationship of functional diversity and information elaboration.

> **Hypothesis 1.** Diversity mindset moderates the relationship of functional diversity with information elaboration such that this relationship is more strongly positive when diversity mindset is higher.

As we outlined in the previous, team information elaboration is the mechanism through which team informational resources are used to produce innovation outcomes (van Knippenberg, 2017). Functional diversity can therefore be expected to have an indirect effect, contingent on diversity mindset (as per Hypothesis 1), on team innovation mediated by information elaboration. Accordingly, we propose moderated mediation, such that functional diversity is more strongly positively related to team innovation, mediated by information elaboration, when diversity mindset is higher.

Hypothesis 2. Functional diversity has an indirect effect on team innovation moderated by diversity mindset such that this relationship, mediated by information elaboration, is stronger with higher diversity mindset.

Boundary spanning scouting and diversity mindset

Functional diversity concerns team informational resources brought to the team through team composition. Boundary spanning scouting is a complementary way in which teams may acquire informational resources – from outside of the team. The focus of boundary spanning scouting (Ancona & Caldwell, 1992) is on gathering knowledge, information, and perspectives to share within the team. As research on information elaboration shows, the fact that information is brought to the team does not mean it will be discussed and integrated into team output (Hoever et al., 2012; van Ginkel & van Knippenberg, 2008; Winquist & Larson Jr., 1998). We can therefore expect that boundary spanning scouting does not automatically result in team information elaboration. This means that for team boundary spanning scouting, just like for team functional diversity, there is value in considering moderation in its relationship with information elaboration.

Here too, we propose that core to this moderation is the team's focus on information elaboration to draw on the team's diverse informational resources. In the analysis advanced by Ancona and Caldwell (1992), boundary spanning scouting is intended to acquire information that the team does not hold internally. In effect, this can be understood through the lens of diversity: The function of boundary spanning scouting is to complement the team's internal informational resources with novel external information. This is well-aligned with the notion that the integration of diverse information and perspectives stimulates team innovation. Accordingly, we argue that diversity mindset with its focus on elaboration to benefit from the team's diverse informational resources is also an important moderating influence for the relationship of boundary spanning scouting with information elaboration.

In this respect, it is important to recognize that diversity mindset is understood to concern a diversity attribute of interest. This makes it important to address why this may also apply to boundary spanning scouting. Key here is that diversity mindset theory sees an understanding of the informational diversity associated with a diversity attribute of interest as the communality for diversity mindsets concerning all diversity attributes (van Knippenberg et al., 2013; cf. van Knippenberg et al., 2004). That is, diversity mindset invites a focus on tapping into the informational diversity of the team. Although this may be triggered by the team's diversity on the one attribute more than on the other, the focus is not on the diversity attribute itself, but on the underlying informational diversity. Accordingly, diversity mindset for a given diversity attribute – the one we consider in this study is functional background – invites a focus on the team's underlying informational diversity associated with functional diversity. Such a focus on exchanging, discussing, and integrating diverse information and perspectives can be expected to result in the team also benefiting more from the information brought to the team through boundary spanning scouting. This may hold especially because members that engaged in boundary spanning scouting likely would present those insights through the lens of their own functional perspectives. Thus, drawing on the same overarching theory identifying a moderating role for diversity mindset in the relationship between team informational resources and elaboration that provides the rationale for Hypothesis 1, we would predict the following:

Hypothesis 3. Diversity mindset moderates the relationship of boundary spanning scouting with information elaboration such that this relationship is more strongly positive when diversity mindset is higher.

As per the knowledge integration perspective on team innovation (van Knippenberg, 2017), information elaboration is the key driver of team innovation. Following the same rationale in which Hypothesis 2 followed from Hypothesis 1, the rationale for Hypothesis 3 can thus be extended to predict the moderated mediation for boundary spanning scouting. Because boundary spanning scouting more strongly stimulates information elaboration with higher diversity mindset, we can expect a stronger indirect effect of boundary spanning scouting, mediated by information elaboration, on team innovation with higher diversity mindset.

Hypothesis 4. Boundary spanning scouting has an indirect effect on team innovation moderated by diversity mindset such that this relationship mediated by information elaboration is stronger with higher diversity mindset.

METHODS

Sample and procedure

To test our hypotheses, we collected three-wave multi-source data from 215 knowledge-intensive teams from organizations across various industries, ranging from information technology, pharmaceutical, consultancy and banking, health care, construction, transportation, manufacturing, real estate, trades, media, hospitality, education, government, and so forth. We contacted these teams via students in an organizational behaviour course in the MBA program in a leading business school in central China. Students' assistance in reaching out to eligible teams was voluntary and the data collection process was closely coordinated and monitored for quality control by one of the authors. Every student was asked to collect data from two to four teams in organizations that they had good contacts with. In the end, 223 teams were contacted and every respondent was rewarded 20 Yuan (approximately 3 US Dollars) for participating in the research.

The data collection took 6 weeks in total (including T_0). The surveys consisted of two waves of team member surveys (T_1 and T_2) and one wave of team leader survey (T_3) with a two-week interval between every two waves. This multi-wave and multi-source design was used to reduce common method biases among focal variables (Gray et al., 2023; Podsakoff et al., 2003). One week before the launch of the first survey (T_0), an introductory meeting was organized for the MBA students in which the class instructor (one coauthor) explained the research purpose, team selection criteria, and research procedure. Specifically, teams needed to consist of three to 10 persons (excluding the team leader), have common goals, perform organizationally relevant tasks, have interdependence among each other, interact with each other, and share the feeling of being a team (Kozlowski & Ilgen, 2006). We also provided the students with a list of team types that were considered teams that performed knowledge work, including R&D teams, professional teams, designs teams, project teams, sales and marketing teams, management teams, service teams (especially service teams for complex industrial products), and others. We specifically asked students not to include production teams, administrative teams, or teams that performed physical labour tasks. Within 2 days after this introductory meeting, the students obtained an agreement from team leaders to participate in the research as well as the names and contact information of members in the team. In this way, we could calculate team size and further administer the two team member surveys. We also asked the team leader to indicate what type of their team was. In all the three waves of surveys, it was emphasized that participation in the research was voluntary and that all the information would be kept anonymous and confidential and used only for scientific purposes. All the surveys were translated from English to Chinese by the two authors who were proficient in both languages and experienced in research. The surveys were administered online to respondents, and they confirmed their consent for participation before starting the survey.

Our sampling strategy is a convenience sampling strategy that is used to collect a sufficient sample in team research (e.g., Gray et al., 2023). Although convenience sampling may render a concern with generalizability (Passmore & Baker, 2005), this concern can be addressed and sample representativeness assured by selecting teams from various industries, organizations, and team types (Landers & Behrend, 2015), as is the case in our study.

In the first member survey (T_1), we assessed boundary spanning scouting and members' team tenure, educational background, educational level, and demographic information (gender, age, organizational tenure). In the second member survey (T_2), we assessed diversity mindset and information elaboration (the moderator and mediator) and members' job title and responsibilities in the team (to calculate functional diversity). In the leader survey (T_3), we assessed team innovation, and team longevity (how long the team had been working together), and leader educational background, educational level, team tenure, leader tenure, and demographic information (gender, age, organizational tenure).

Among the 223 teams we contacted (i.e., 223 leaders and 933 members), 215 teams (i.e., 215 leaders and 898 members) responded to our surveys with valid answers (96.41% response rate). The within-team response rates of the three waves were 91.73%, 88%, and 100%, respectively. Among the 898 members, 44 did not respond to the first survey (4.9% missing data rate) and 78 not to the second survey (8.7% missing data rate). The missing data rates were below 10% and thus considered to be negligible (Cheema, 2014; Dong & Peng, 2013).

In addition to the missing data rate, the randomness of nonresponse is an issue, because non-random nonresponse can bias model estimation and thus requires imputations (Ployhart & Vandenberg, 2010). To check the randomness of nonresponse, we performed Little's Test of Missing Completely at Random (TMCR) for the study variables (at the item level for a multi-item measure). We excluded job title (as a measure of functional background), because TMCR cannot test categorical variables. A non-significant result meant that missing data was random and unrelated to the study variables. Our results showed non-significant Little's TMCR results, that is, $\chi^2 = 39.01$, df $= 36$, $p = .34$, suggesting missing completely at random. In this case (i.e., negligible and random nonresponse), we could have used listwise deletion and deleted teams with missing responses at T_1 or T_2. However, to preserve as many observations as possible in the sample, we imputed missing data with expectancy-maximization (EM) estimates. Imputing EM estimates is considered a superior way to deal with missing data compared pairwise and listwise deletion, as it generates estimates as if the missing were completely at random. In this way, researchers do not only keep all the possible observations in the sample (for multi-wave and longitudinal studies) but also can remedy non-random nonresponse and reduce model estimation bias (even though the latter is not necessary in our case, Cheema, 2014; Newman, 2003).

Among the 215 teams, 51 teams (23.72%) are sales and marketing teams, 48 (22.33%) professional teams, 38 (17.67%) project teams, 31 (14.42%) R&D teams, 20 (9.3%) management teams, 12 (5.58%) design teams, 12 (5.58%) service teams, and 3 (1.40%) other teams. Excluding the team leaders, the average team size is 4.18 persons (s.d. = 1.41) and average team longevity is 2.88 years (s.d. = 2.57).

Measures

Functional diversity was measured by calculating Blau's (1977) index for members' functional backgrounds. To measure functional background, we asked each participant (a) to indicate what job title they had and (b) to provide a short description of what his or her major responsibilities were in the team. Three trained research assistants (RAs) coded each job title based on the 2015 National Guide of Job Classifications of China. This national guide classifies jobs in China into eight categories, 75 subcategories, 1481 professions, and 2670 positions. We used the 75 subcategories in the coding to ensure that respondents' functional backgrounds were distinctive enough from each other yet not totally idiosyncratic so that team functional background diversity would not be a mere reflection of team size. After RA coding, the two Chinese native authors on the team re-checked the coding and re-coded wherever needed in consultation with respondents' descriptions of their responsibilities in the team.

Boundary spanning scouting was measured with Ancona and Caldwell's (1992) three-item scale at T_1. We asked the team members to what extent they agreed (1 = strongly disagree; 7 = strongly agree) with the following items 'our team finds out what competing companies or teams are doing on similar projects', 'our team scans the environment inside and outside the company for ideas or expertise', and 'our team collects information or ideas from individuals outside the team'. Cronbach $a = .87$. mean $r$wg = .79, ICC (1) = .13, ICC (2) = .38.

Diversity mindset was measured with van Knippenberg et al.'s (2019) nine-item scale at T_2. This scale was developed following diversity mindset theory (van Knippenberg et al., 2013) to capture the goal of benefiting from diversity in non-routine, explorative performance through information elaboration. Diversity mindset is understood to be specific to a diversity attribute and team, such that the focus in operationalizing it should reference respondents' own team and the diversity attribute of interest (van Knippenberg et al., 2013). Across two studies in the Netherlands and China, van Knippenberg et al. (2019) showed that their scale could reference cultural diversity as well as gender diversity with similar psychometric results. In the present study, we extrapolated from their findings and used their scale to measure diversity mindset referencing functional diversity with such items as 'to perform well, in our team, we should focus on learning new insights from members with different functional backgrounds', and 'to perform well, in our team, we should invest in the exchange of insights inspired by our different functional backgrounds'. Responses were on 7-point scales ranging from 1 = strongly disagree to 7 = strongly agree. Cronbach $a = .94$, mean $r$wg = .95, ICC (1) = .14, ICC (2) = .40. As outlined in the Introduction, we worked on the assumption that our operationalization captured members' underlying focus on informational diversity in their team (i.e., which would include information brought to the team through boundary spanning). This is because boundary spanning scouting cannot be referenced in diversity mindset measurement like a diversity attribute such as functional background can.

Information elaboration was measured with van Knippenberg et al.' (2019) six-item scale at T_2. This scale was developed following the categorization-elaboration model definition of team information elaboration (van Knippenberg et al., 2004) and validated across a study in The Netherlands and China. We asked the team members to what extent they agreed (1 = strongly disagree; 7 = strongly agree) with such items as 'in our team, members complement each other by sharing their knowledge', and 'in our team, we discuss alternatives before making decisions'. Cronbach $a = .93$, mean $r$wg = .91, ICC (1) = .14, ICC (2) = .41.

Team innovation was measured with four items from De Dreu's (2006) scale at T_3. We asked the team leaders to what extent they agreed (1 = strongly disagree; 7 = strongly agree) with such items as 'team members often produce new services, methods, or procedures', and 'this is an innovative team'. Cronbach alpha = .91.

Control variables included in the models are team size (how many members are in this team), team longevity (how long the team has been working together), and team type (which type this team is). Team size and team longevity were obtained from the team leaders. team types were controlled by constructing seven dummy variables (1 = yes, and 0 = no) for the seven team types: Team A means R&D teams, Team B means professional teams, Team C means designs teams, Team D means project teams, Team E means sales and marketing teams, Team F means management teams, and Team G means service teams.

RESULTS

We performed team-level confirmatory factor analysis (CFA) with Mplus 7.4 (Muthén & Muthén, 1998–2015). We first aggregated the items of boundary spanning scouting, diversity mindset, and information elaboration to the team level, and we then tested the team-level CFA. The four-factor model (boundary spanning scouting, diversity mindset, information elaboration, and team innovation) had good model fit, $\chi^2 = 382.63$, df = 203, CFI = .96, TLI = .95, RMSEA = .06, SRMR = .03. The three-factor model (combining boundary spanning scouting and diversity mindset) shows a lower model fit, $\chi^2 = 633.93$, df = 206, CFI = .90, TLI = .89, RMSEA = .10, SRMR = .06. The two-factor model (combining boundary spanning scouting, diversity mindset and information elaboration) shows an unacceptable model fit, $\chi^2 = 970.17$, df = 208, CFI = .83, TLI = .81, RMSEA = .13, SRMR = .07. The one-factor model (combining all four variables into one factor) shows the worst model fit, $\chi^2 = 1546.07$, df = 209, CFI = .70, TLI = .67, RMSEA = .17, SRMR = .12.

Descriptive statistics and correlations for the study variables are presented in Table 1. As shown, the intercorrelations between boundary spanning scouting, diversity mindset, and information elaboration are higher than is desirable. CFA results showed however that the four-factor model had a good fit and that this fit was better than alternative models. This suggested that these high correlations are the result of the strong covariation of the variables and do not reflect a measurement problem. Therefore, we concluded it was appropriate to proceed to hypothesis testing despite these high intercorrelations.

Hypothesis tests

We tested the full model using Mplus 7.4. The regression analysis results are presented in Table 2. The first column of tests statistics concerns the prediction of team information elaboration for the test of Hypothesis 1 and 3. The variables making up the functional diversity × diversity mindset and boundary spanning scouting × diversity mindset interaction terms were mean-centred before their product was computed. The second column of tests statistics concerns the prediction of team innovation, for which mainly the relationship between information elaboration and innovation is of interest as an element in (but not a test of) Hypothesis 2 and 4.

Hypothesis 1 states that diversity mindset moderates the relationship of functional diversity with information elaboration such that this relationship is more strongly positive when diversity mindset is higher. The results showed that the interaction between functional diversity and diversity mindset on information elaboration was not significant, $b = 0.03$, *ns*. Thus, Hypothesis 1 is not supported.

Hypothesis 2 states that functional diversity has an indirect effect on team innovation moderated by diversity mindset such that this relationship, mediated by information elaboration, is stronger with higher diversity mindset. The results showed that information elaboration was positively related to team innovation, $b = 0.66$, $p < .05$. Not surprisingly, given that Hypothesis 1 was not supported, bootstrapping the moderated indirect effect of functional diversity showed that there was no significant moderated mediation effect, 0.02, 95% CI [−.26, .31]. Hypothesis 2 thus was not supported.

Hypothesis 3 states that diversity mindset moderates the relationship of boundary spanning scouting with information elaboration such that this relationship is more strongly positive when diversity mindset is higher. The results showed that the effect of boundary spanning scouting on information elaboration was significant, $b = 0.30$, $p < .001$, and so was that of diversity mindset, $b = 0.64$, $p < .001$. The interaction between boundary spanning scouting and diversity mindset on information elaboration was significant, $b = 0.11$, $p < .05$, thus supporting Hypothesis 3. Figure 2 presents the interaction through simple slopes for diversity mindset one standard deviation below and above the mean, As shown, the positive relationship between boundary spanning scouting and elaboration was stronger, $b = 0.35$, $p < .001$, when diversity mindset was high than when diversity mindset was low, $b = 0.24$, $p < .001$ (the difference between the two slopes was significant, $p < .05$).

Hypothesis 4 states that boundary spanning scouting has an indirect effect on team innovation moderated by diversity mindset such that this relationship mediated by information elaboration is stronger with higher diversity mindset. Bootstrapping of the moderated indirect effect of boundary spanning scouting showed that this effect was significant, 0.08, 95% CI [.01, .22]. As predicted, there was a positive indirect effect that was stronger with high diversity mindset, 0.23, 95% CI (.05, .48), when compared with low diversity mindset, 0.16, 95% CI (.03, .35). Thus, Hypothesis 4 was supported.

TABLE 1 Descriptive statistics and correlations for study variables.

Variables	*M*	SD	1	2	3	4	5	6	7	8	9	10	11	12	13
1. Team A	.14	.35													
2. Team B	.22	.42	−.22***												
3. Team C	.06	.23	−.10	−.13											
4. Team D	.18	.38	−.19**	−.25***	−.11										
5. Team E	.24	.43	−.23***	−.30***	−.14*	−.26***									
6. Team F	.09	.29	−.13	−.17*	−.08	−.15*	−.18**								
7. Team G	.06	.23	−.10	−.13	−.06	−.11	−.14*	−.08							
8. Team size	4.18	1.41	.01	.20**	−.06	−.10	.02	−.05	−.09						
9. Team longevity	2.88	2.57	.08	.00	.01	−.04	−.08	−.01	.01	−.11					
10. Functional diversity	.32	.31	−.02	.00	−.08	.03	.06	−.06	−.02	−.03	−.04				
11. Boundary spanning scouting	5.59	.69	−.01	−.06	−.08	.11	.10	−.02	−.14*	.00	.02	.11			
12. Diversity mindset	5.78	.53	−.09	.00	−.18*	.05	.07	.04	.04	.04	−.03	.13	.66***		
13. Information elaboration	5.66	.59	−.14*	.02	−.14*	.09	.05	.01	−.01	.05	.03	.11	.69***	.77***	
14. Team innovation	5.44	1.10	.09	.01	.08	−.10	−.15*	.04	.12	.09	.07	−.12	.14*	.06	.20**

Note: $N = 215$ teams, $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$. Team A = R&D teams, Team B = professional teams, Team C = designs teams, Team D = project teams, Team E = sales and marketing teams, Team F = management teams, Team G = service teams.

TABLE 2 Results for hypothesis testing.

	Testing H1 and H3	
Variable	**Information elaboration**	**Team innovation**
Intercept	2.44***	4.87***
Team A	−.58***	.15
Team B	−.42***	−.18
Team C	−.51***	.25
Team D	−.42***	−.47
Team E	−.50***	−.49
Team F	−.47***	−.01
Team G	−.45***	.52
Team size	.02	.07
Team longevity	.004	.02
Functional diversity (FD)	−.004	−.41
Boundary spanning scouting (BSS)	.30***	.20
Diversity mindset (DM)	.64***	−.52
Information elaboration		.66*
FD × DM	.03	−.41
BSS × DM	.11*	.05
R^2	.69***	.16**
Testing H2		
Indirect effect at low DM	−.01 (95%CI [−.22, .16])	
Indirect effect at high DM	.01 (95%CI [−.16, .19])	
Differences of indirect effects	.02 (95%CI [−.26, .31])	
Testing H4		
Indirect effect at low DM	.16 (95%CI [.03, .35])	
Indirect effect at high DM	.23 (95%CI [.05, .48])	
Differences of indirect effects	.08 (95%CI [.01, .22])	

Note: $N = 215$ teams, bootstrapping $n = 20{,}000$, $^*p < .05$, $^{**}p < .01$, $^{***}p < .001$. Team A = R&D teams, Team B = professional teams, Team C = designs teams, Team D = project teams, Team E = sales and marketing teams, Team F = management teams, Team G = service teams.

DISCUSSION

The aim of the current study was to contribute to the knowledge integration perspective on team innovation by bridging team diversity and team boundary spanning approaches to team informational resources. We argued that these approaches can be bridged by recognizing that the influences of functional diversity and boundary spanning scouting on team innovation can both be linked to information elaboration as a mediating process and to the moderating influence of diversity mindset. To achieve this integration, we drew upon team diversity theory that emphasizes the importance of information elaboration (van Knippenberg et al., 2004) and diversity mindset (van Knippenberg et al., 2013). Somewhat ironically, then, our hypotheses were supported for boundary spanning scouting but not for functional diversity.

Against the backdrop of other evidence that information elaboration (or related concepts like knowledge sharing) mediates moderated relationships of different forms of informational diversity with team creativity and innovation (Cheung et al., 2016; Drach-Zahevy & Somech, 2001; Hoever et al., 2012, 2018; Qu & Liu, 2017; Somech & Drach-Zahavy, 2007; Zhang, 2016), it seems warranted to consider why these hypotheses were not supported in the current study – as we do in the following – rather than

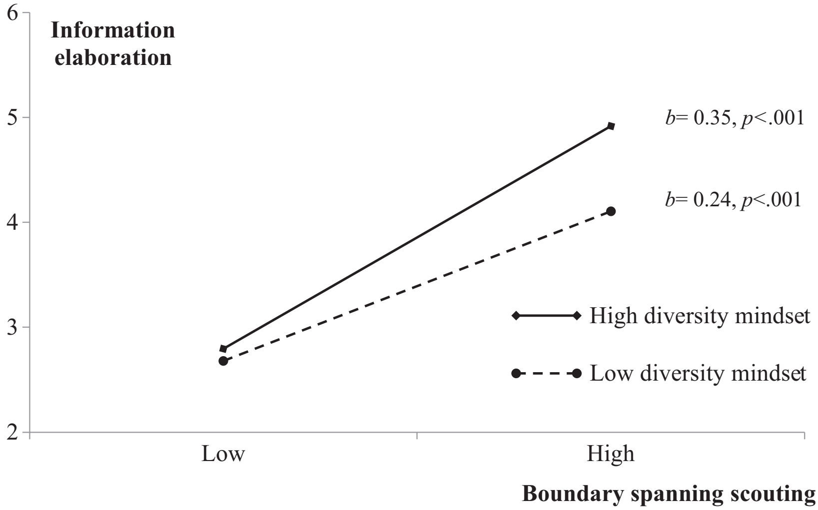

FIGURE 2 The interaction of boundary spanning scouting and diversity mindset on information elaboration.

to simply discard these hypotheses. More important from the perspective of bridging diversity and boundary spanning approaches is the finding that information elaboration mediated the indirect effect of boundary spanning scouting on team innovation, contingent on diversity mindset. This provides firmer ground for the further development of the integration of these approaches to team informational resources and team innovation, as we also consider in the following.

Theoretical implications

The starting point for our analysis was the knowledge integration perspective on team innovation and its recognition of the value of team informational resources (van Knippenberg, 2017). This led us to identify the communalities between team functional diversity and boundary spanning scouting as influences on team innovation that can both be understood from the knowledge integration perspective: both (i) reflect team informational resources, that (ii) require information elaboration to result in team innovation, which (iii) is contingent on moderating influences stimulating the team to tap into its diverse informational resources. These recognitions allow us to link both functional diversity and boundary spanning scouting to team innovation through the same moderated (diversity mindset) mediation (information elaboration).

The fact that our hypotheses developed from team diversity theory are supported for boundary spanning scouting suggests that we may achieve further integration of the two team informational resource approaches by drawing more broadly on diversity theory and research. The categorization-elaboration model (van Knippenberg et al., 2004) proposes that factors associated with team member motivation and ability for information elaboration moderate the effect of team diversity on elaboration. In the current study, we considered this issue from the perspective of diversity mindset, members' shared understanding of the importance of information elaboration to utilize diversity (van Knippenberg et al., 2013). Motivational influences are not limited to diversity mindset, however. As theory and research outlines, member motivation for information elaboration may for instance flow from a desire to form accurate judgement that can have a dispositional (Kearney et al., 2009) and a situational (Scholten et al., 2007) basis, from a learning goal orientation (Nederveen Pieterse et al., 2013), and from member openness to experience (Homan et al., 2008). Our intention here is not to give a review of such influences (see e.g., Guillaume et al., 2017), but to note that the present analysis suggests that such moderating influences should also hold for the effect of boundary spanning scouting on information elaboration – and thus also for its indirect effect on team innovation as an outcome of elaboration.

Putting this proposition to the test in future research is valuable, because it would develop a broader basis for an integrative analysis of team informational resources as an influence on team innovation. Further establishing that the communality of team diversity and boundary spanning scouting as reflecting team informational resources justifies such integrative propositions would also provide a firmer basis to apply the same logic to teams' external social network. Social networks are understood to be a source of information and perspectives that may stimulate not only individual creativity and innovation (Zhou et al., 2009) but also team creativity and innovation (Chen, 2009; Han et al., 2014; Kowlaser & Barnard, 2016; Perry-Smith & Shalley, 2014). The team's external network can thus be understood to concern team informational resources just like team diversity and team boundary spanning. It is therefore a small step to suggest that our perspective bridging team diversity and team boundary spanning approaches on team innovation can be extended to also include the team external network approach. Here too, the logic is that having informational resources – external network ties – cannot be equated to using these resources in team information elaboration. Importantly, the influence of the external network can also not be reduced to a more proximal influence of team boundary spanning scouting as the activity of drawing on the external network. Just as boundary spanning can vary independent of team diversity, the composition (e.g., size, diversity; Baer, 2010) of the external network can vary independent of boundary spanning scouting (as well as independent of team diversity). The team's external network thus is an influence on team innovation to consider in its own right – as an influence that may be mobilized through information elaboration.

We can only speculate why our hypotheses concerning functional diversity are not supported. In view of (i) meta-analytic evidence for the heterogeneous effects (i.e., suggesting moderating influences) of functional diversity on team innovation (van Dijk et al., 2012), (ii) multiple studies establishing moderation in the diversity-innovation relationship as well as mediation by elaboration (van Knippenberg & Hoever, 2021), and (iii) the current support for Hypothesis 3 and 4 that underscores that diversity mindset has the hypothesized moderating role in the relationship between team informational resources and elaboration and innovation, it seems not warranted to dismiss Hypothesis 1 and 2. Probably, the question that is more on target is why the predicted effects were too weak in the current analysis to establish.

A first thing to note is that there is reason to expect boundary spanning scouting effects to be stronger than diversity effects, because boundary spanning to scout for information is explicitly focused on bringing information that is perceived to be task-relevant to the team. Team members engaged in boundary spanning scouting likely also actively pursue sharing this information with the team. This does not mean that information is always shared, and sharing information does not necessarily lead to integration (van Ginkel & van Knippenberg, 2008; Winquist & Larson Jr., 1998), but this may mean that boundary spanning scouting compared with functional diversity may make elaboration more likely (as per also the main effect of boundary spanning) and thus have a lower 'threshold' for the moderating effect of diversity mindset.

Diversity, in contrast, is an aspect of team composition and not an activity focused on bringing information to the team. As research on transactive memory (shared knowledge about the distribution of knowledge within the team) shows, team members often are not fully aware of their functional diversity and draw more on the team's diversity the better their transactive memory (Richter et al., 2012). In that sense, boundary spanning scouting likely makes the associated informational resources more easily available for information elaboration than functional background diversity, because it concerns actions to bring novel information to the team. Thus, one possibility to consider is that the effects of functional background diversity may have been weaker compared than those of boundary spanning scouting because they need more moderating conditions (such as better developed transactive memory) to bring them out.

A related possibility is that we have a restriction of range issue with functional diversity. Our theories typically, but implicitly, assume the full range of the variables we study, but it seems reasonable to assume that for almost all our studies the reality is that they can only tap into a more restricted part of that range. We are not aware of objectives standards to determine whether the range of a variable in a given study is large enough to observe hypothesized effect when they exist in the population, but one possibility is that a larger range of functional diversity is required to establish the predicted relationships. We do not have the data to explore these issues, however, and can only leave this as a consideration for future research.

Practical implications

The present findings underscore the value of the knowledge integration perspective on team innovation both in showing that team informational resources are positively related to team innovation and in showing that the information elaboration required to benefit from those resources does not automatically follow from the availability of the resources. Even boundary spanning to scout for external information, an activity explicitly focused on bringing task-relevant information and perspectives to the team, does not guarantee that the team engages in more information elaboration. In addition to further emphasizing that team innovation benefits from team informational resources – and that boundary spanning scouting is a way to provide such resources – the current findings thus point to the contingent nature of team informational resources effects on team innovation. In doing so, they emphasize the value of developing teams' diversity mindset.

Our study does not speak to the antecedents of diversity mindset. Accordingly, in this respect we can only draw on diversity mindset theory and note that this comes with the caveat that this does not directly follow from study findings. Diversity mindset theory identified leadership to stimulate the development of diversity mindset as a particularly relevant way to shape team diversity mindset (van Knippenberg et al., 2013; van Knippenberg & van Ginkel, 2022). The rationale here is that diversity mindset is team-specific and thus best developed 'locally', in the context of the team and teamwork. Team leadership in that sense provides a local influence on such development. Diversity mindset theory proposes that leaders can influence team diversity mindset by (i) engaging the team in conversations about its diversity in which the leader advocates an understanding of diversity as an informational resource from which team performance can benefit through information elaboration, (ii) coaching the team in the elaboration process, and (iii) guiding the team in collective reflection on this experience to further, develop the team's diversity mindset. This comes with the caveat that the current study does not concern leadership but is to illustrate that diversity mindset theory also points to actionable ways in which the innovation benefits of diversity mindset can be realized.

Limitations and future directions

Our study design has its strengths in relying on multi-wave and multi-source data and a research model in which no single relationship concerns single-source main effect relationships (i.e., interaction effects cannot be explained by single source bias; Evans, 1985). Even so, our design is not without its limitations. An obvious limitation is that we rely on subjective ratings for variables that, in principle at least, could also be assessed more objectively in that boundary spanning scouting, information elaboration, and team innovation are all behavioural and therefore potentially open to more objective measurement (diversity mindset is team cognition and inevitably relies on subjective measurement). The current subjective measurement may introduce 'noise' that makes it more difficult to identify true 'signal', for instance by inviting inflated correlations between study variables. This also seems to be an issue in the current study in which, despite the clear support for our measurement model, intercorrelations were substantially higher than they would ideally be. Future research that would address these problems would be valuable. This could for instance be done by using more objective measures of team innovation (cf. West & Anderson, 1996) and by gathering external ratings of boundary spanning scouting to reduce the common source issue.

Another obvious limitation is that even though our theory is causal, our data are correlational and thus mute in matters of causality. In this respect, it is good to note that there is experimental research establishing moderated indirect effects of functional background diversity via information elaboration to team creativity (Hoever et al., 2012, 2018), which adds plausibility to the causality implied in our research model. Obviously, however, causal evidence in other studies does not establish causality in the current study. Research further developing the integration of different approaches to team informational resources would therefore add particular value if it provided (field) experimental evidence.

No study can prove generalizability beyond the context in which it is conducted. From experience we know, however, that studies outside of 'Western' countries are more likely to invite questions about whether findings would generalize beyond the cultural context in which the study was conducted. There is a clear case to argue that all studies regardless of where they are conducted should be expected to consider the generalizability of findings to the same degree and that questions of generalizability should not be prioritized for research originating outside of Western countries (Avery et al., 2022). We fully agree with that argument and here simply note that the evidence for the knowledge integration perspective that identifies team information elaboration as the core process driving team innovation derives from studies across different countries and continents and suggests no country-specific deviations from the general pattern (Hülsheger et al., 2009; van Dijk et al., 2012; van Knippenberg, 2017).

CONCLUSION

Knowledge integration lies at the core of team innovation and this makes concepts reflecting team informational resources particularly valuable to study in developing our understanding of team innovation. As we hope to establish with the present study, the recognition of the communality between different approaches to team informational resources sets the stage for the development of more integrative theory about team innovation. The present study is only one step in that direction. It is also a less than perfect step, ironically, in its inability to find support for its predictions drawn from diversity theory for team diversity. But we believe it is important in paving the way to developing a more integrated team informational resource approach to the study of team innovation.

AUTHOR CONTRIBUTIONS

Daan van Knippenberg: Writing – original draft; conceptualization. **Jia Li:** Formal analysis; writing – review and editing; methodology. **Yidong Tu:** Writing – review and editing; methodology; formal analysis; funding acquisition; investigation.

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CONFLICT OF INTERESTS STATEMENT

The authors state no conflict of interest regarding this research.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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REFERENCES

Ancona, D. G., & Caldwell, D. F. (1992). Demography and design: Predictors of new product team performance. *Organization Science*, *33*, 321–341.

Anderson, N., Potočnik, K., & Zhou, J. (2014). Innovation and creativity in organizations: A state-of-the-science review, prospective commentary, and guiding framework. *Journal of Management*, *40*, 1297–1333.

Avery, D. R. B., Darren, K., Dumas, T. L., George, E., Joshi, A., Loyd, D. L., van Knippenberg, D., Wang, M., & Xu, H. (2022). Racial biases in the publication process: Exploring expressions and solutions. *Journal of Management*, *48*, 7–16.

Baer, M. (2010). The strength-of-weak-ties perspective on creativity: A comprehensive examination and extension. *Journal of Applied Psychology*, *95*, 592–601.

Blau, P. (1977). *Inequality and heterogeneity*. Free Press.

Cheema, J. R. (2014). A review of missing data handling methods in education research. *Review of Educational Research*, *84*, 487–508.

Chen, M. H. (2009). Guanxi networks and creativity in Taiwanese project teams. *Creativity and Innovation Management*, *18*, 269–277.

Cheung, S. Y., Gong, Y., Wang, M., Zhou, L., & Shi, J. (2016). When and how does functional diversity influence team innovation? The mediating role of knowledge sharing and the moderating role of affect-based trust in a team. *Human Relations*, *69*, 1507–1531.

Cronin, M. A., Stouten, J., & van Knippenberg, D. (2021). The theory crisis in management research: Solving the right problem. *Academy of Management Review*, *46*, 667–683.

De Dreu, C. K. W. (2006). When too little or too much hurts: Evidence for a curvilinear relationship between task conflict and innovation in teams. *Journal of Management*, *32*, 83–107.

De Dreu, C. K. W., Nijstad, B. A., & van Knippenberg, D. (2008). Motivated information processing in group judgment and decision making. *Personality and Social Psychology Review*, *12*, 22–49.

Dong, Y., & Peng, C.-Y. J. (2013). Principled missing data methods for researchers. *Springerplus*, *2*, 222.

Drach-Zahevy, A., & Somech, A. (2001). Understanding team innovation: The role of team processes and structures. *Group Dynamics*, *5*, 111–123.

Ely, R. J., & Thomas, D. A. (2001). Cultural diversity at work: The effects of diversity perspectives on work group processes and outcomes. *Administrative Science Quarterly*, *46*, 229–273.

Evans, M. G. (1985). A Monte Carlo study of the effects of correlated method variance in moderated regression analysis. *Organizational Behavior and Human Decision Processes*, *36*, 305–323.

Gray, S. M., Bunderson, J. S., van der Vegt, G. S., Rink, F., & Gedik, Y. (2023). Leveraging knowledge diversity in hierarchically differentiated teams: The critical role of hierarchy stability. *Academy of Management Journal*, *66*, 462–488.

Guillaume, Y. R. F., Dawson, J. F., Otaye-Ebede, L., Woods, S. A., & West, M. A. (2017). Harnessing demographic differences in organizations: What moderates the effects of workplace diversity? *Journal of Organizational Behavior*, *38*, 267–303.

Han, J., Han, J., & Brass, D. J. (2014). Human capital diversity in the creation of social capital for team creativity. *Journal of Organizational Behavior*, *35*, 54–71.

Hargadon, A. B., & Bechky, B. A. (2006). When collections of creatives become creative collectives: A field study of problem solving at work. *Organization Science*, *17*, 484–500.

Hoever, I. J., van Knippenberg, D., van Ginkel, W. P., & Barkema, H. G. (2012). Fostering team creativity: Perspective taking as key to unlocking diversity's potential. *Journal of Applied Psychology*, *97*, 982–996.

Hoever, I. J., Zhou, J., & van Knippenberg, D. (2018). Different strokes for different teams: The contingent effects of positive and negative feedback on the creativity of informationally homogeneous and diverse teams. *Academy of Management Journal*, *61*, 2159–2181.

Homan, A. C., Hollenbeck, J. R., Humphrey, S. E., van Knippenberg, D., Ilgen, D. R., & Van Kleef, G. A. (2008). Facing differences with an open mind: Openness to experience, salience of intra-group differences, and performance of diverse work groups. *Academy of Management Journal*, *51*, 1204–1222.

Howell, J. M., & Shea, C. M. (2006). Effects of champion behavior, team potency, and external communication activities on predicting team performance. *Group & Organization Management*, *31*, 180–211.

Hülsheger, U. R., Anderson, N., & Salgado, J. F. (2009). Team-level predictors of innovation at work: A comprehensive meat-analysis spanning three decades of research. *Journal of Applied Psychology*, *94*, 1128–1145.

Kearney, E., Gebert, D., & Voelpel, S. C. (2009). When and how diversity benefits teams: The importance of team members' need for cognition. *Academy of Management Journal*, *52*, 581–598.

Kooij-De Bode, H. J. M., van Knippenberg, D., & van Ginkel, W. P. (2008). Ethnic diversity and distributed information in group decision making: The importance of information elaboration. *Group Dynamics*, *12*, 307–320.

Kowlaser, K., & Barnard, H. (2016). Tie breadth, tie strength and the location of ties: The value of ties inside an emerging MNC to team innovation. *International Journal of Innovation Management*, *20*, 1–31.

Kozlowski, S. W., & Ilgen, D. R. (2006). Enhancing the effectiveness of work groups and teams. *Psychological Science in the Public Interest*, *7*, 77–124.

Landers, R. N., & Behrend, T. S. (2015). An inconvenient truth: Arbitrary distinctions between organizational, Mechanical Turk, and other convenience samples. *Industrial and Organizational Psychology*, *8*, 142–164.

Morgeson, F. P., & Hoffman, D. A. (1999). The structure and function of collective constructs: Implications for multilevel research and theory development. *Academy of Management Review*, *24*, 249–265.

Muthén, L. K., & Muthén, B. O. (1998–2015). *Mplus user's guide (7th ed.)*. Muthén and Muthén.

Nederveen Pieterse, A., van Knippenberg, D., & van Dierendonck, D. (2013). Cultural diversity and team performance: The role of team member goal orientation. *Academy of Management Journal*, *56*, 782–804.

Newell, A., & Simon, H. A. (1972). *Human problem solving*. Prentice Hall.

Newman, D. A. (2003). Longitudinal modeling with randomly and systematically missing data: A simulation of ad hoc, maximum likelihood, and multiple imputation techniques. *Organizational Research Methods*, *6*, 328–362.

Passmore, D. L., & Baker, R. M. (2005). Sampling strategies and power analysis. In R. A. Swanson & E. F. Holten (Eds.), *Research in organizations: Foundations and methods of inquiry* (pp.45-56). San Francisco: Berrett–Koehler.

Perry-Smith, J. E., & Shalley, C. E. (2014). A social composition view of team creativity: The role of member nationality-heterogeneous ties outside of the team. *Organization Science*, *25*, 1434–1452.

Phelps, C., Heidl, R., & Wadhwa, A. (2012). Knowledge, networks, and knowledge networks: A review and research agenda. *Journal of Management*, *38*, 1115–1166.

Ployhart, R. E., & Vandenberg, R. J. (2010). Longitudinal research: The theory, design, and analysis of change. *Journal of Management*, *36*, 94–120.

Podsakoff, P. M., MacKenzie, S. B., Lee, J. Y., & Podsakoff, N. P. (2003). Common method biases in behavioral research: A critical review of the literature and recommended remedies. *Journal of Applied Psychology*, *88*, 879–903.

Qu, X., & Liu, X. (2017). Informational faultlines, integrative capability, and team creativity. *Group & Organization Management*, *42*, 767–791.

Richter, A., Hirst, G., van Knippenberg, D., & Baer, M. (2012). Creative self-efficacy and individual creativity in teams: Cross-level interactions with team informational resources. *Journal of Applied Psychology*, *97*, 1282–1290.

Salas, E., & Fiore, S. M. (2004). *Team cognition: Understanding the factors that drive process and performance*. APA.

Scholten, L., van Knippenberg, D., Nijstad, B. A., & De Dreu, C. K. W. (2007). Motivated information processing and group decision making: Effects of process accountability on information sharing and decision quality. *Journal of Experimental Social Psychology*, *43*, 539–552.

Schotter, A. P. J., Mudambi, R., Doz, Y. L., & Gaur, A. (2017). Boundary spanning in global organizations. *Journal of Management Studies*, *54*, 403–421.

Shalley, C. E., Hitt, M. A., & Zhou, J. (2015). *Oxford handbook of creativity, innovation, and entrepreneurship*. Oxford University Press.

Somech, A., & Drach-Zahavy, A. (2007). Schools as team-based organizations: A structure-process-outcomes approach. *Group Dynamics*, *11*, 305–320.

van Dijk, H., van Engen, M. L., & van Knippenberg, D. (2012). Defying conventional wisdom: A meta-analytical examination of the differences between demographic and job-related diversity relationships with performance. *Organizational Behavior and Human Decision Processes*, *119*, 38–53.

van Ginkel, W. P., & van Knippenberg, D. (2008). Group information elaboration and group decision making: The role of shared task representations. *Organizational Behavior and Human Decision Processes*, *105*, 82–97.

van Ginkel, W. P., & van Knippenberg, D. (2009). Knowledge about the distribution of information and group decision making: When and why does it work? *Organizational Behavior and Human Decision Processes*, *108*, 218–229.

van Knippenberg, D. (2017). Team innovation. *Annual Review of Organizational Psychology and Organizational Behavior*, *4*, 211–233.

van Knippenberg, D., De Dreu, C. K. W., & Homan, A. C. (2004). Work group diversity and group performance: An integrative model and research agenda. *Journal of Applied Psychology*, *89*, 1008–1022.

van Knippenberg, D., & Hoever, I. J. (2021). Team innovation and team creativity. In J. Zhou & E. Rouse (Eds.), *Handbook of research on creativity and innovation* (pp. 49–66). Edward Elgar.

van Knippenberg, D., & van Ginkel, W. P. (2022). A diversity mindset perspective on inclusive leadership. *Group & Organization Management*, *47*, 779–797.

van Knippenberg, D., van Ginkel, W. P., & Homan, A. C. (2013). Diversity mindsets and the performance of diverse teams. *Organizational Behavior and Human Decision Processes*, *121*, 183–193.

van Knippenberg, D., van Ginkel, W. P., Stam, D. A., Giessner, S. R., van Balen, T., & Ghorbani, M. (2019). Diversity mindsets, gender dissimilarity, and performance. *Paper presented at the Annual Meeting of the Academy of Management, Boston*.

West, M. A., & Anderson, N. R. (1996). Innovation in top management teams. *Journal of Applied Psychology*, *81*, 680–693.

Winquist, J. R., & Larson, J. R., Jr. (1998). Information pooling: When it impacts group decision making. *Journal of Personality and Social Psychology*, *74*, 371–377.

Zhang, Y. (2016). Functional diversity and group creativity: The role of group longevity. *The Journal of Applied Behavioral Science*, *52*, 97–123.

Zhou, J., Shin, S. J., Brass, D. J., Choi, J., & Zhang, Z. X. (2009). Social networks, personal values, and creativity: Evidence for curvilinear and interaction effects. *Journal of Applied Psychology*, *94*, 1544–1552.

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